



Incident handler's journal

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this journal as a way to log the key takeaways about the different cybersecurity tools or concepts you encounter in this course.

Date: July 14, 2024	Entry: #1
Description	A ransomware incident affecting a small U.S. healthcare clinic. The attack began with a phishing email containing a malicious attachment. Once executed, ransomware encrypted critical files, including patient medical records, leading to a complete shutdown of business operations. The response involved two main phases: 1. Detection and Analysis – Identified the ransomware through employee reports and the ransom note. 2. Containment, Eradication, and Recovery – The clinic isolated affected systems and sought external technical assistance, as they lacked sufficient internal resources to recover independently.
Tool(s) used	None
The 5 W's	● Who caused the incident? An organized group of unethical hackers specializing in targeting healthcare organizations. ● What happened? Attackers used phishing emails to deliver ransomware, which encrypted critical patient data and operational files, causing major disruption to healthcare services. ● When did the incident occur?

	Tuesday morning at approximately 9:00 a.m. ● Where did the incident happen? A small healthcare clinic providing primary care services in the United States. ● Why did the incident happen? The attackers exploited weak email security controls and lack of employee awareness training. The primary motivation appears to be financial, as evidenced by the ransom demand for the decryption key.
Additional notes	This incident highlights the severe impact ransomware can have on healthcare organizations, especially regarding patient data availability and confidentiality. Key improvement areas include enhanced phishing protection, regular offline backups, and incident response planning.

Date: July 15, 2024	Entry: #2
Description	Investigated a suspicious file download alert. The file was a password-protected spreadsheet delivered via phishing email. After the employee opened the file, a malicious payload was executed. I generated a SHA256 hash of the suspicious file and analyzed it using VirusTotal to identify Indicators of Compromise (IOCs).
Tool(s) used	☐ VirusTotal ☐ SHA256 hash function
The 5 W's	● Who caused the incident? Likely an attacker or threat actor using phishing as the delivery method to target employees. ● What happened?

	An employee received a phishing email with a malicious spreadsheet attachment. After entering the provided password and opening the file, malware was executed on the endpoint. ● When did the incident occur? Alert received at approximately 1:20 p.m. ● Where did the incident happen? On an employee workstation at a financial services company (Level One SOC). ● Why did the incident happen? The employee was tricked by a social engineering attack (phishing email with password-protected attachment), highlighting gaps in security awareness and email filtering controls.
Additional notes	The SHA256 hash was submitted to VirusTotal, which returned multiple IOCs (Indicators of Compromise) including malicious verdicts from several security vendors. Key findings included associated IP addresses, domains, and behavioral indicators consistent with malware execution. This incident reinforces the need for better email security controls and user training on handling suspicious attachments.

Date: July 20, 2024	Entry: #3
Description	Reviewed the final incident report of a major data breach that occurred at a mid-sized retail company. The breach involved unauthorized access to customer PII and financial information affecting approximately 50,000 records. The review helped me understand the full incident lifecycle from initial compromise to response and remediation.
Tool(s) used	☐ Final Incident Report document

The 5 W's	● Who caused the incident? An external attacker who exploited a vulnerability in the company's e-commerce web application. ● What happened? The attacker used a forced browsing attack by modifying order numbers in purchase confirmation URLs, allowing access to customer transaction data. This led to the theft and attempted extortion of customer PII and financial information. ● When did the incident occur? Initial suspicious email on December 22, 2022, with the main breach occurring on December 28, 2022, at approximately 7:20 p.m. PT. ● Where did the incident happen? On the company's e-commerce web application and associated backend systems. ● Why did the incident happen? Due to a web application vulnerability (insufficient access controls on purchase confirmation pages) combined with inadequate monitoring of unusual access patterns.
Additional notes	The company responded by investigating the breach, notifying affected customers, and offering identity protection services. Key recommendations included regular vulnerability scanning, penetration testing, improved access control mechanisms, and better URL validation. This case reinforced the importance of secure web application development, continuous monitoring, and having a strong incident response plan in place.